Research Seminar:
Data Analysis in Finance

Group 17

How to Measure the Liquidity of Cryptocurrency Markets?

A replication of Brauneis, Mestel, Riordan & Theissen (2021) on independently collected Tardis.dev exchange data, with our own robustness extension

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Moscow, 2026

PROBLEM

The question we ask

Crypto exchanges publish no shared order-book history. Can cheap measures from prices and volume tell us what trading really costs?

- **Paper** Brauneis et al. race 8 low-frequency proxies against 4 order-book benchmarks (BTC, ETH; 2017–2019).
- **Our replication** Does their ranking hold on independently recorded exchange data from 2019?
- **Our extension** Is the agreement more than shared volatility? Does a calibrated proxy measure cost levels on later dates?

"Does the cheap proxy see the real cost of trading?"



A high correlation is not the same as the right cost level.

Brauneis, A., Mestel, R., Riordan, R., Theissen, E. (2021). How to measure the liquidity of cryptocurrency markets? Journal of Banking & Finance 124, 106041 (SSRN version, March 2020).

THE PAPER · AUTHORS' RESULTS

What the paper finds

Brauneis et al. (2020): Bitcoin, hourly, 16 Dec 2017 – 16 Jun 2019. All numbers on this slide are the authors'.

0.59–0.84

AR's correlation with the four benchmarks: the highest of all eight proxies

≈ 0

Amihud ratio: virtually uncorrelated with every benchmark

0.35–0.72

TX and $Vol correlate positively with costs: more trading, higher costs

66.67%

of hours in which AR orders the three venues exactly like the quoted spread

The authors' recommendation

Timing and comparing venues → AR

CS is second for QS, ES and PI. A principal-component mix of all proxies does not beat AR.

Cost levels → Kyle and Amihud

Smallest level errors (Table 5); AR overstates the level. Kyle tracks the roundtrip cost (0.69) but not the spreads.

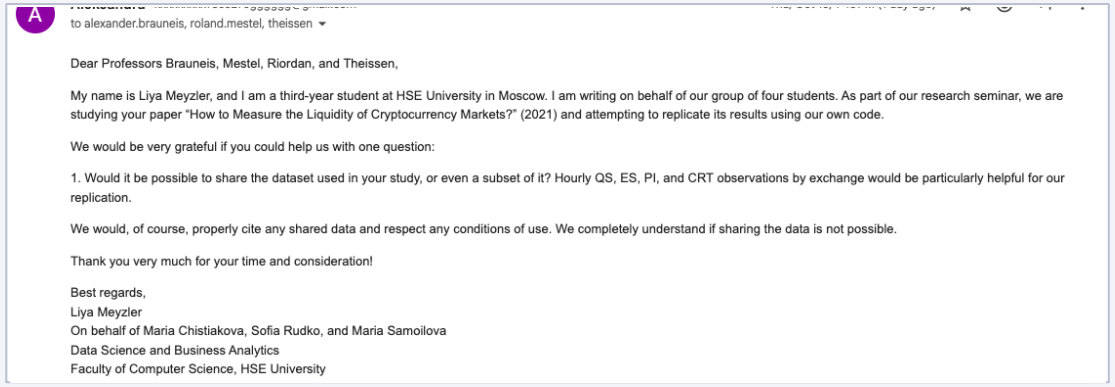
Source: Brauneis et al. (SSRN 2020), Figure 1, Tables 5–6 and Section 4; published 2021 in the Journal of Banking & Finance.

DATA

Our data: collected independently from Tardis.dev

We asked the authors for their dataset and have had no reply, so we built our own sample from recorded exchange feeds.

1 We asked the authors



to alexander.brauneis, roland.mestel, theissen

Dear Professors Brauneis, Mestel, Riordan, and Theissen,

My name is Liya Meyzler, and I am a third-year student at HSE University in Moscow. I am writing on behalf of our group of four students. As part of our research seminar, we are studying your paper "How to Measure the Liquidity of Cryptocurrency Markets?" (2021) and attempting to replicate its results using our own code.

We would be very grateful if you could help us with one question:

1. Would it be possible to share the dataset used in your study, or even a subset of it? Hourly QS, ES, PI, and CRT observations by exchange would be particularly helpful for our replication.

We would, of course, properly cite any shared data and respect any conditions of use. We completely understand if sharing the data is not possible.

Thank you very much for your time and consideration!

Best regards,
Liya Meyzler
On behalf of Maria Chistiakova, Sofia Rudko, and Maria Samoilova
Data Science and Business Analytics
Faculty of Computer Science, HSE University

Email to all four authors, sent on behalf of the group. No reply so far.

2 So we collected the data ourselves

7 days

1st UTC day of each month, June–December 2019

6 markets

Bitfinex, Bitstamp, Coinbase × BTC/USD, ETH/USD

126 files

trades, best quotes, 25-level books · 2.9 GiB

807 / 1,008

market-hours pass the paper's 48-of-60 rule

Not the authors' dataset: independent recordings of the exchanges' live WebSocket feeds, not their REST-API snapshots. Only 1 June 2019 lies inside the paper's period, so this is an out-of-sample check.

Tardis.dev historical CSV datasets: trades, quotes and book_snapshot_25. Raw and minute-level files stay local (Tardis.dev terms of service).

DATA

Same markets, different data and design

Our results are an independent check of the paper's findings, not a copy of its tables.

	Brauneis et al. (paper)	Our study
Period	16 Dec 2017 – 16 Jun 2019	7 days, Jun–Dec 2019
Data source	authors' REST-API polling	Tardis.dev feed recordings
Order book	≤ 50 levels, irregular snapshots	25 levels, 10-second grid
Frequency	hourly, daily, 15-day	hourly
Hourly intervals	≈ 9,500	807 valid of 1,008
Evaluation	correlations, quantiles, PCA, regimes, errors, rankings	correlations, regimes, errors, rankings + our extension

Valid hours per market

	BTC	ETH
Bitfinex	164	122
Bitstamp	167	**28**
Coinbase	163	163

Bitstamp ETH is thin: read with care. 1 Aug 2019 has recording gaps on Bitfinex and in Coinbase trades.

1 Data and sample
Maria Chistiakova

2 Eight proxies
Sofia Rudko

3 Benchmarks and comparison
Liya Meyzler

4 Robustness and extension
Maria Samoilova

Paper: about 9,500 hourly intervals after the same 80% (48-of-60 minutes) coverage rule. Notebook Sections 2.7 and 3.6.

PROXIES · PART 2

Eight proxies, four ideas

All eight need only minute open, high, low, close, number of trades and USD volume.

ACTIVITY

TX, $Vol

How much people trade: trades and USD volume per minute.

high = more liquid

PRICE MOVE PER DOLLAR

Amihud, Kyle

How far the price moves for each dollar traded.

high = less liquid

BID–ASK BOUNCE

Roll p, Roll r

Trades flip between bid and ask, so price changes partly undo each other.

high = less liquid

HIGH–LOW RANGE

CS, AR

A minute's high, low and close contain about half a spread (Corwin–Schultz, Abdi–Ranaldo).

high = less liquid

Not interchangeable: AR, CS and Roll r are in bp like the quoted spread; Amihud is per USD, Kyle in $(1/\text{USD})^{1/3}$.

Definitions as in Brauneis et al., Section 2.2.2. Computed per UTC hour from one-minute bars (notebook Section 4).

PROXIES · PART 2

How we computed them, and how we know the code is right

One value per market-hour, from minutes with trades only.

Steps

1. 60,480 one-minute bars: 42 market-days × 1,440 minutes
2. One value per UTC hour, using only minutes with trades
3. Keep the hour if ≥ 48 of 60 minutes have trades: 807 of 1,008
4. Pairs only from neighbouring minutes with trades, never across a gap or an hour

Checks

- **Hand example, 4 minutes:** TX 2.25, $Vol 1,250, Roll p 2.83, as on paper
- **Flat price, true spread 5 bp:** AR 5.00, CS 5.00, Roll 5.02
- **Totals:** TX and volume add back up to the hourly totals exactly
- **Recomputation:** proxies match the saved file (max. relative difference $7.7 \cdot 10^{-12}$)



Every proxy is checked before any result is read.

AR, CS and Roll r in bp, like QS; Amihud and Kyle depend on USD units. Notebook Sections 4.1–4.6.

PROXIES · PART 2

AR gets the timing right, but overstates small spreads

Within one minute, the high–low range is mostly price movement, not the spread.

BITFINEX BTC · 164 HOURS · MEDIANS

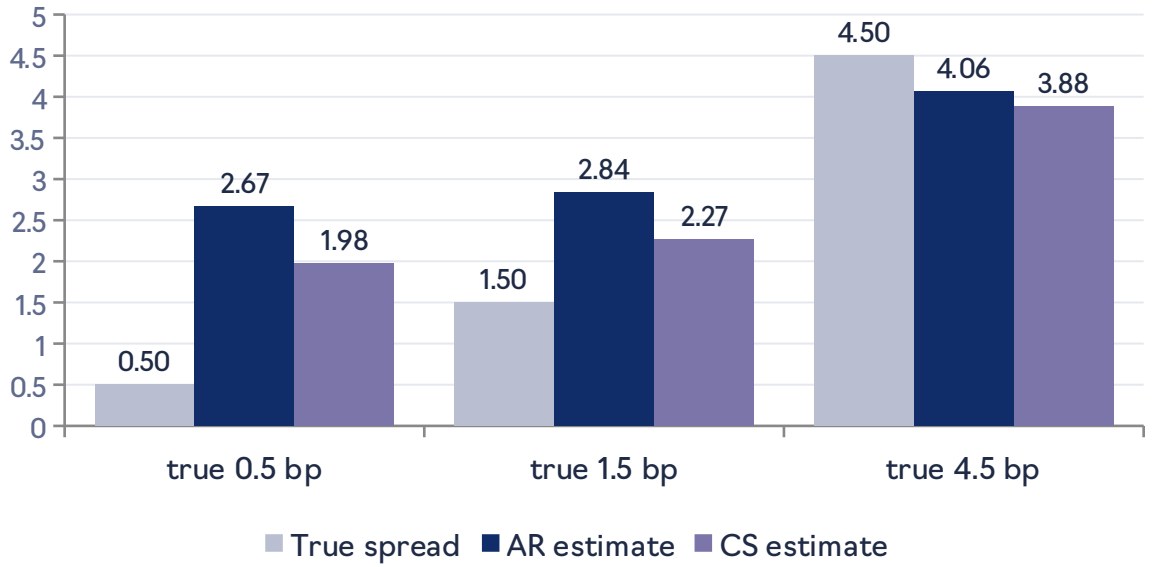
0.52 bp → **1.73 bp**

true quoted spread — AR estimate: more than 3×

59% of Bitfinex BTC hours have Roll = 0 (42–59% on Bitfinex and Coinbase)

29–60% of AR and CS minute pairs are negative and set to 0, as in the paper: an upward push

Simulated Bitfinex-like market (bp)



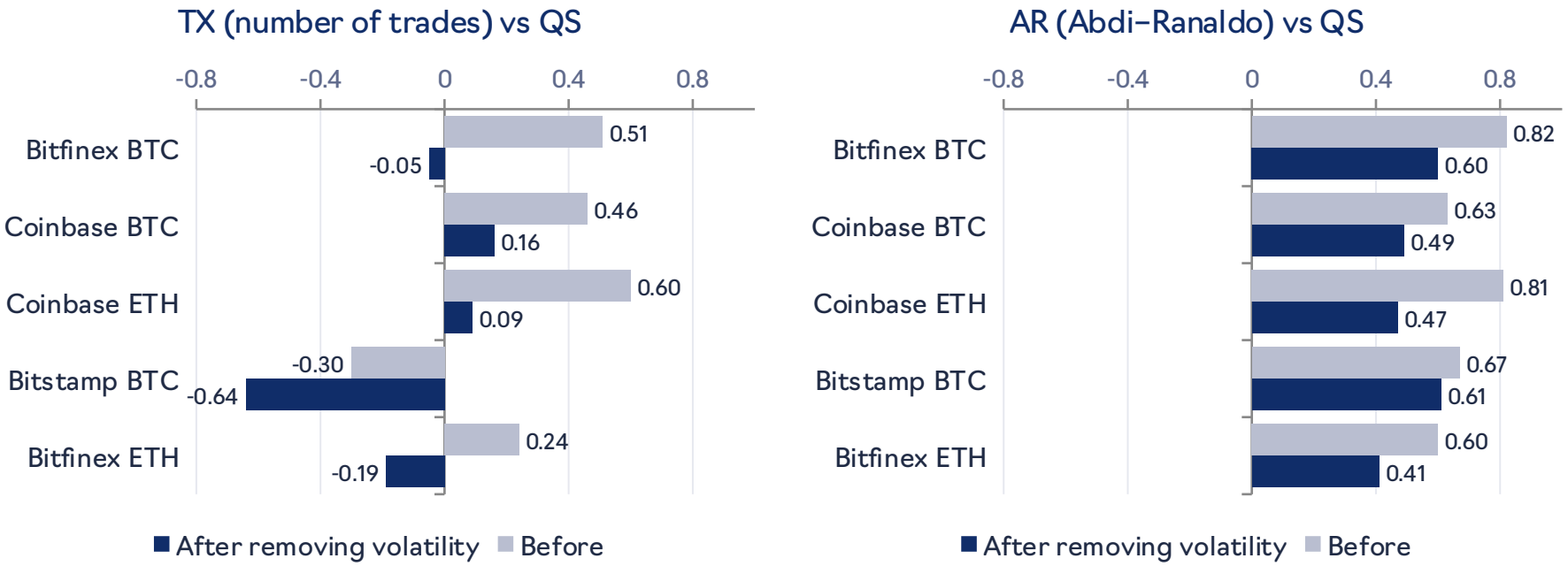
Spread × 9 → AR only × 1.5. Same bias without any data problems.

Medians over valid hours (notebook Sections 4.8–4.9). Simulation: random-walk price, trades at bid or ask, inputs taken from our own medians.

PROXIES · PART 2

Is the paper's “surprising sign” just volatility?

Spearman correlation with QS before and after removing hourly volatility, five larger markets.



SIMULATION

AR −0.03
spread moves on its own

AR 0.66
spread and trading follow volatility

AR −0.01
same, volatility removed

TX's link with QS disappears or turns negative; AR keeps 0.41–0.61. AR sees the spread itself.

Partial Spearman correlation on ranked hourly volatility. Bitstamp ETH (28 hours) not shown. Part 4 repeats this for all four benchmarks with bootstrap intervals.

BENCHMARKS · PART 3

Four benchmarks: what a trade really costs

The “true” liquidity, built from best quotes, trades and 25-level order books on a 10-second grid.

QS

Quoted spread

(ask − bid) / midpoint at every valid grid point

ES

Effective spread

2 × |trade price − midpoint| / midpoint, first trade after a snapshot

PI

Price impact

signed change of the midpoint after the trade

CRT25

Roundtrip cost

buy and sell 32k USD of BTC (18k of ETH) through at most 25 book levels



Our five construction decisions

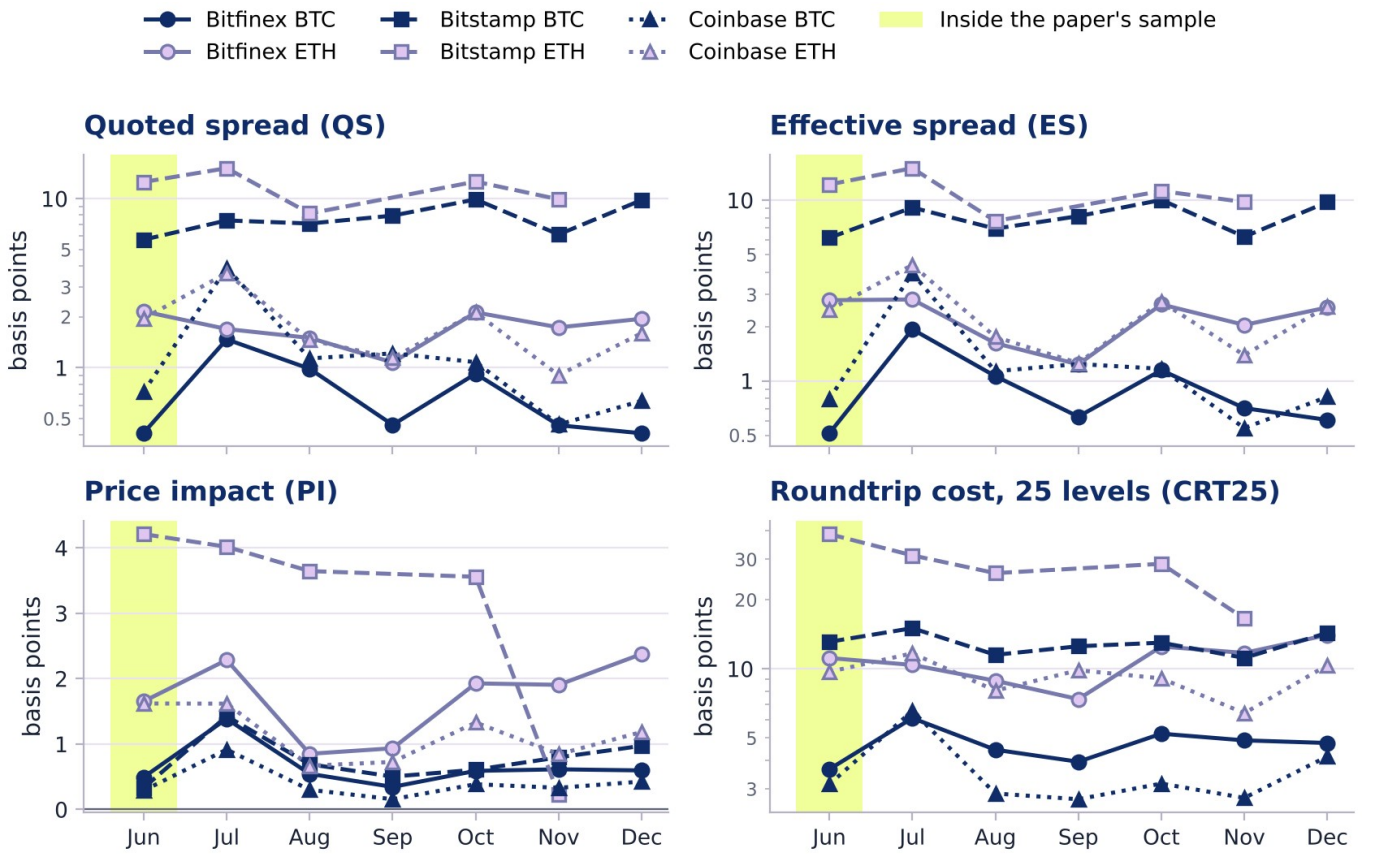
- Trades matched to quotes on the same clock (capture time)
- Quote at most 60 s old (30 s as a check)
- ES as printed (absolute); signed ES as a check
- PI keeps the sign, as the text defines it
- CRT on 25 levels: enough in 99.85% of snapshots

Paper: irregular REST-API snapshots, up to 50 levels. Where the SSRN text and tables disagree (ES, PI, Table 5 units), our choices are documented in notebook Section 2.6.

BENCHMARKS · PART 3

Benchmark levels on the seven dates

The venues line up as in the paper; QS and CRT25 levels are close to it, ES and PI levels are higher.



Close to Table 3

QS Bitfinex BTC: 0.725 bp (paper 0.723). CRT25 BTC on Bitfinex / Bitstamp / Coinbase: 4.72 / 12.96 / 3.64 bp vs 3.72 / 14.30 / 3.41.

Higher than Table 3

ES and PI are 2.4–8.5× the paper's means; our ES exceeds QS in 5 of 6 markets. Likely cause (our hypothesis): on a 10-second grid the quote is several seconds old at the trade.

Coinbase BTC

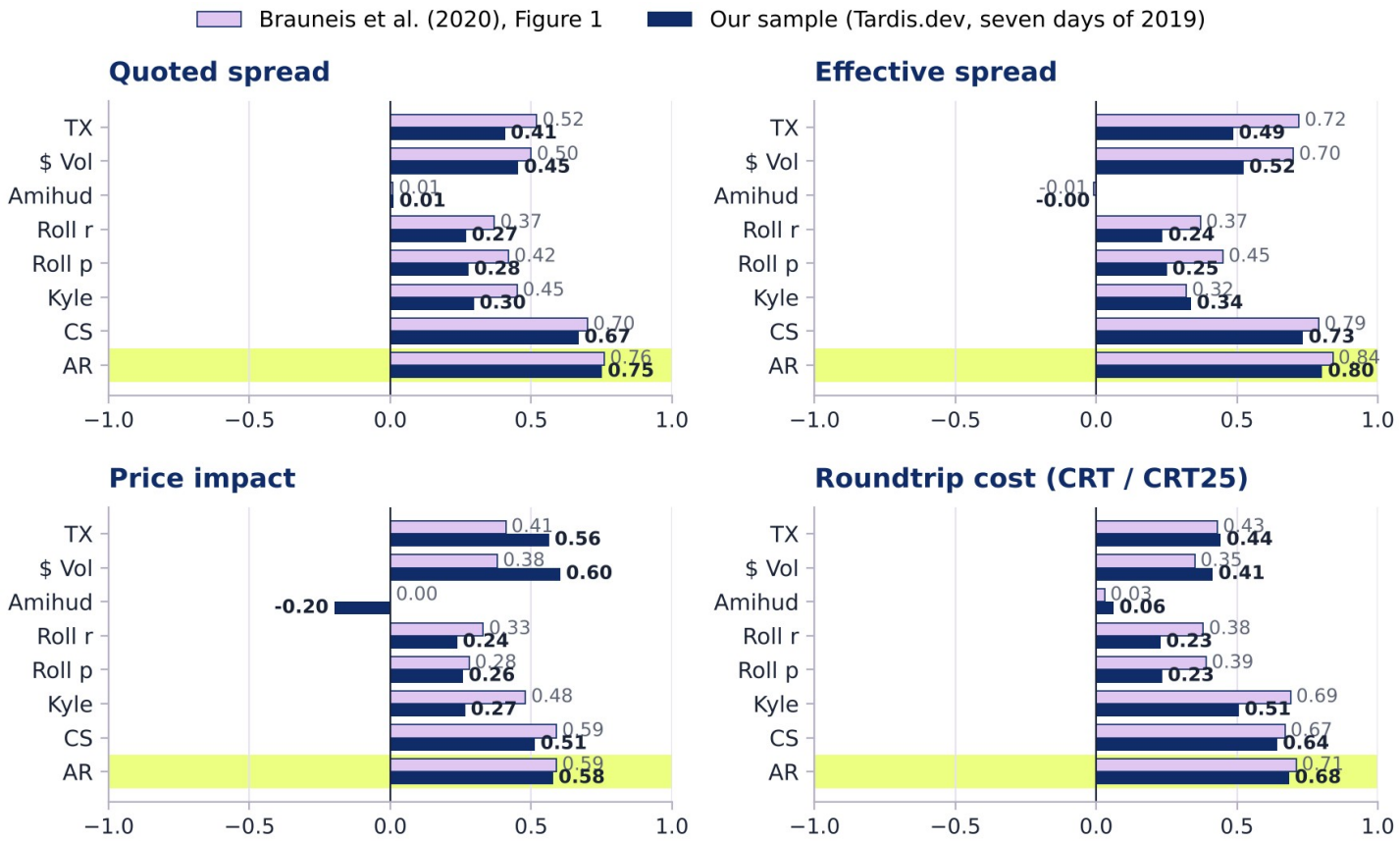
Our QS is about 4× the paper's: 1.31 vs 0.32 bp.

Daily means over valid hours; QS, ES and CRT25 on a log scale. Lime band: 1 June 2019, the only date inside the paper's sample. Paper: Table 3 (BTC), Table 7 (ETH).

RESULTS · PART 3

The main result replicates: AR tracks costs best

Hourly correlations with the four benchmarks, Bitcoin, averaged over the three venues.



Paper (Brauneis et al.) Our sample

AR vs ...	Ours	Paper
Quoted spread	0.75	0.76
Effective spread	0.80	0.84
Roundtrip cost	0.68	0.71
Price impact	0.58	0.59

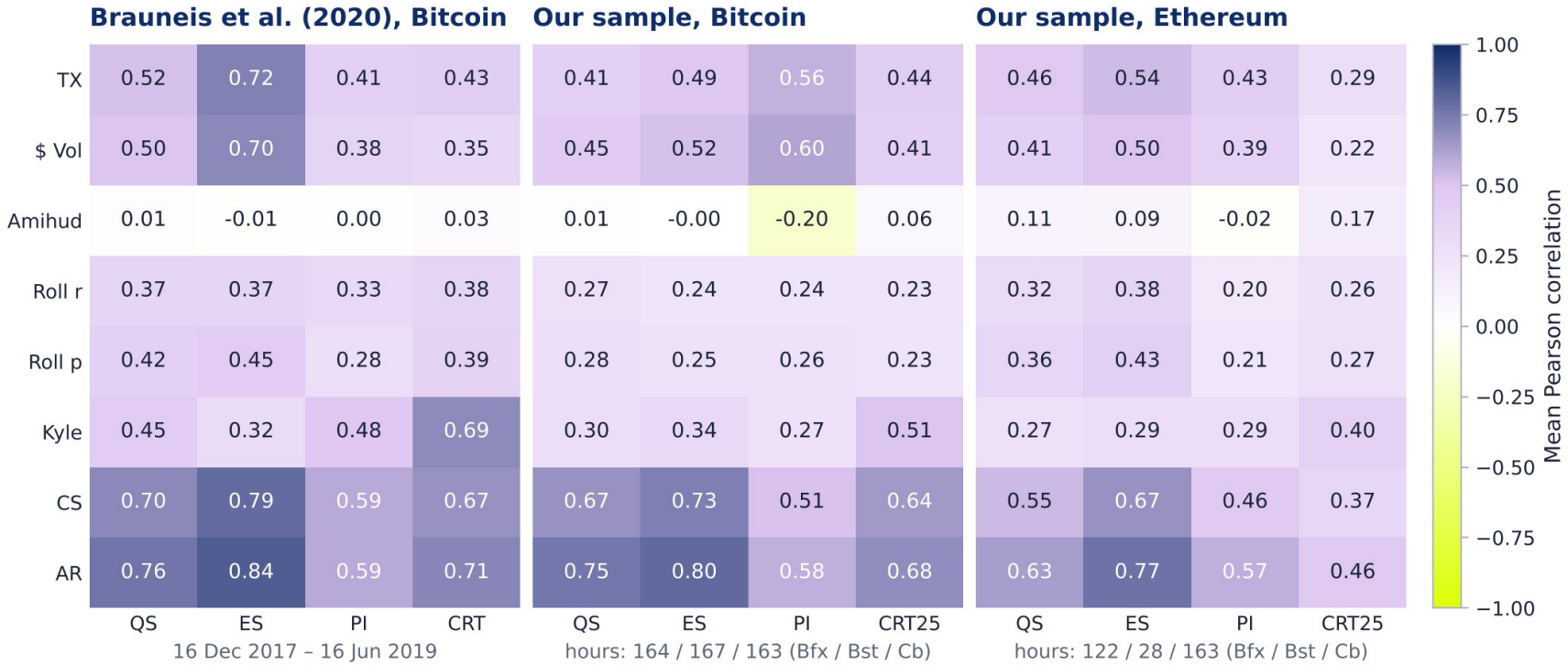
- AR is first for QS, ES and CRT25; for PI, $Vol (0.60) is marginally above AR (0.58).
- Amihud ≈ 0 again; TX and $Vol positive again (0.41–0.60).

Pearson within each venue, averaged over Bitfinex, Bitstamp and Coinbase, as in the paper's Figure 1. Ours: 494 valid BTC hours; CRT25 on 25 levels.

RESULTS · PART 3

Same ordering of proxies, slightly weaker coefficients

Mean correlation of each proxy with each benchmark: the paper (Bitcoin) and our sample (Bitcoin, Ethereum).



Rankings agree more than sizes: 7 days of 2019 against 18 months of 2017–2019.

0.90
correlation of our 32 BTC coefficients with the paper's

0.98
rank agreement of the 8 proxies for QS (ES 0.90, CRT 0.90, PI 0.65)

−0.065
mean difference; ours lower in 24 of 32

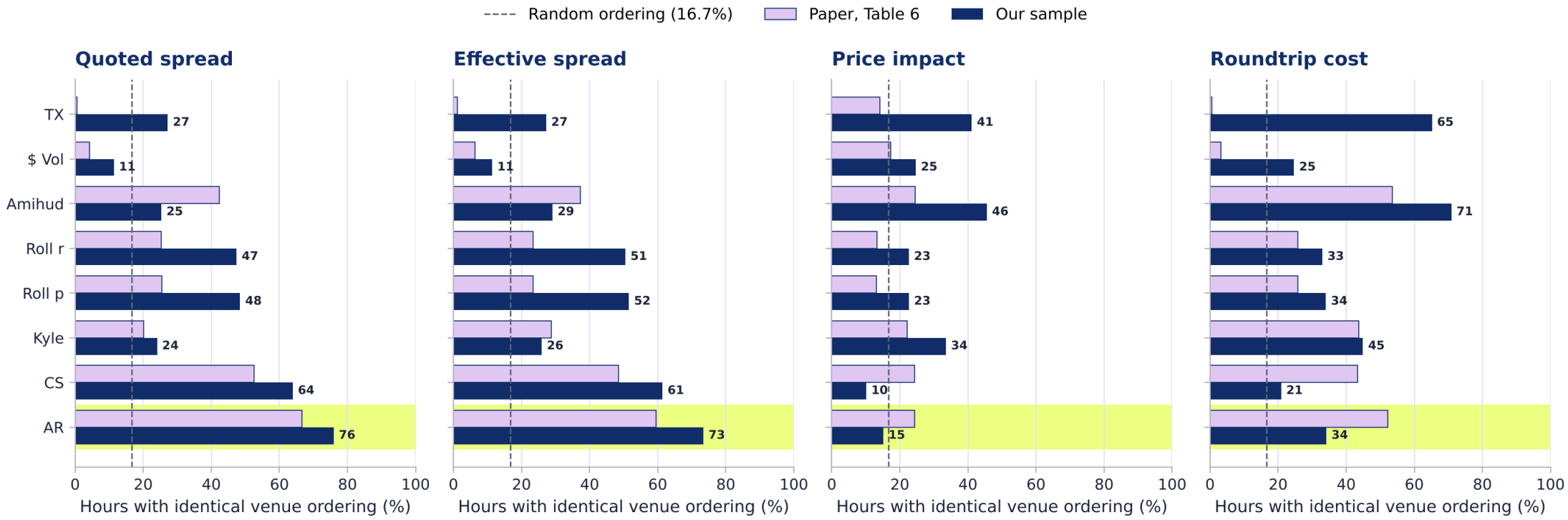
4 of 4
ETH: AR is top for all four benchmarks

Hourly data, valid hours only; colour scale from −1 (lime) through 0 (white) to +1 (navy). The paper's values are the labels printed in its Figure 1.

RESULTS · PART 3

AR also ranks the venues best

How often a proxy orders Bitfinex, Bitstamp and Coinbase exactly like the benchmark (Bitcoin, hourly).



QS: AR 75.9% (paper 66.67%)

ES: AR 73.4% (paper 59.57%)

CRT: Amihud first (70.9%)

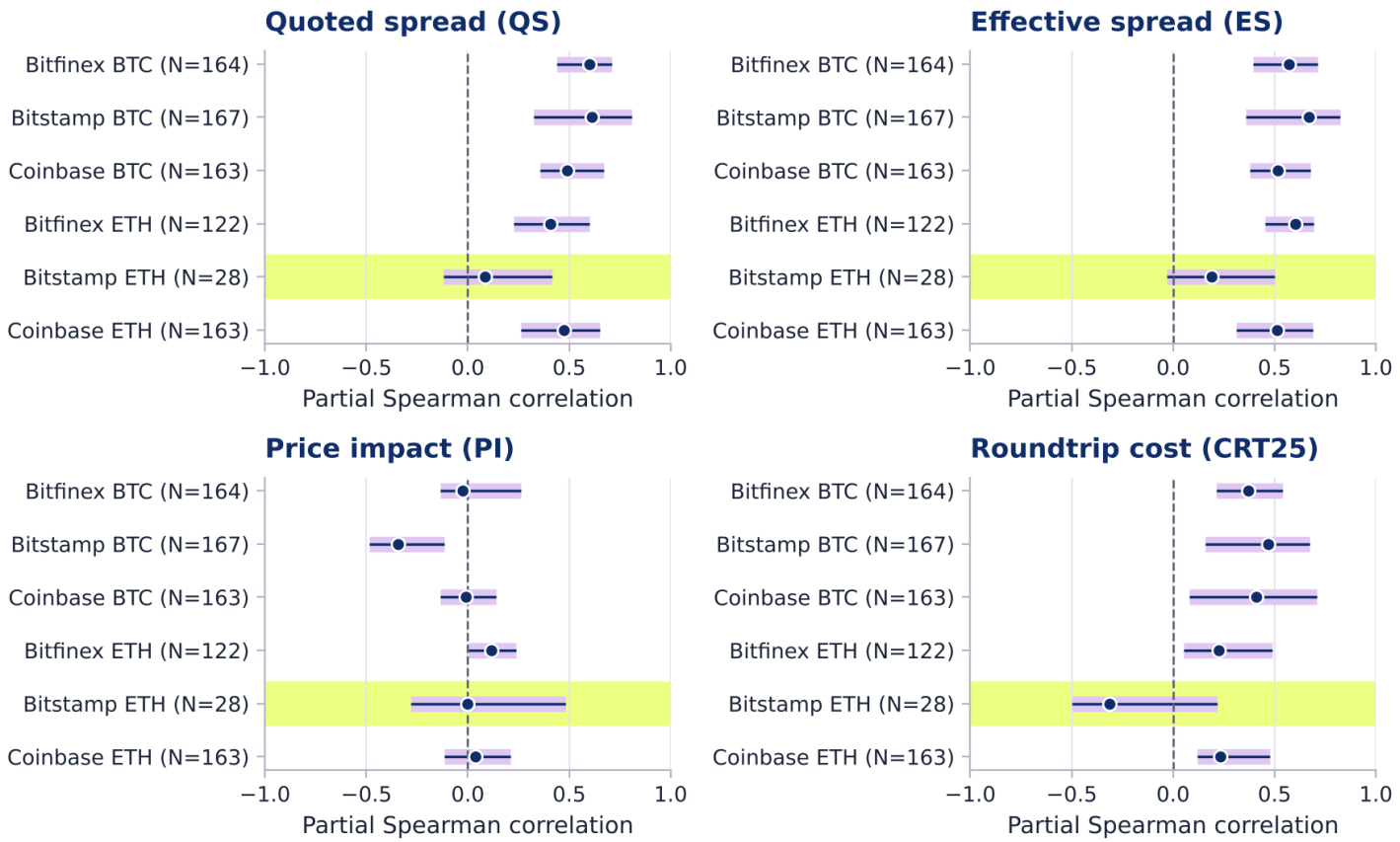
PI: AR 15.2% < random 16.7%

158 hours with all three venues valid; TX and $Vol reversed; ties excluded (Roll: 97 untied hours). ETH QS: AR 61.5% of 26 hours. Paper: Table 6.

EXTENSION · PART 4

Beyond volatility: AR still tracks the spreads

Partial Spearman correlation of AR with each benchmark after removing hourly volatility, with 95% bootstrap intervals.



QS 0.41–0.61 · ES 0.51–0.67
Survive the control; every interval is above zero (lower bounds 0.23–0.45)

CRT25 0.22–0.47
Weaker; lower bounds 0.05–0.21

PI −0.35 to 0.12
Gone; Bitstamp BTC even negative, interval [−0.49, −0.12]

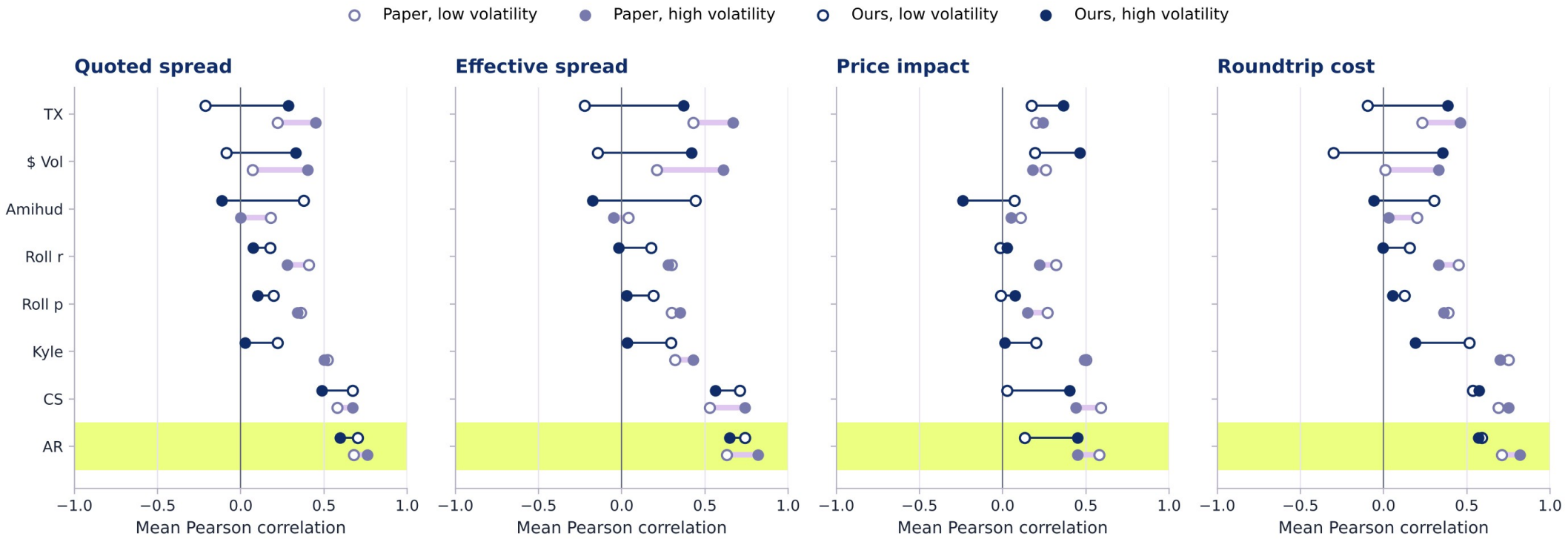
TX and $Vol
Negative partial correlation with CRT25 in all five markets

Ranges over the five larger markets. 999 date-block resamples of the seven dates, same draws for all markets; individual intervals, no multiple-comparison adjustment. Lime band: Bitstamp ETH.

EXTENSION · PART 4

Calm vs volatile hours: the paper's pattern reverses

Bitcoin correlations in the lowest and highest volatility tercile. Open marker: calm hours; filled marker: volatile hours.



AR–QS: 0.70 → 0.60 (paper 0.68 → 0.76)

AR still best for QS and ES in both terciles

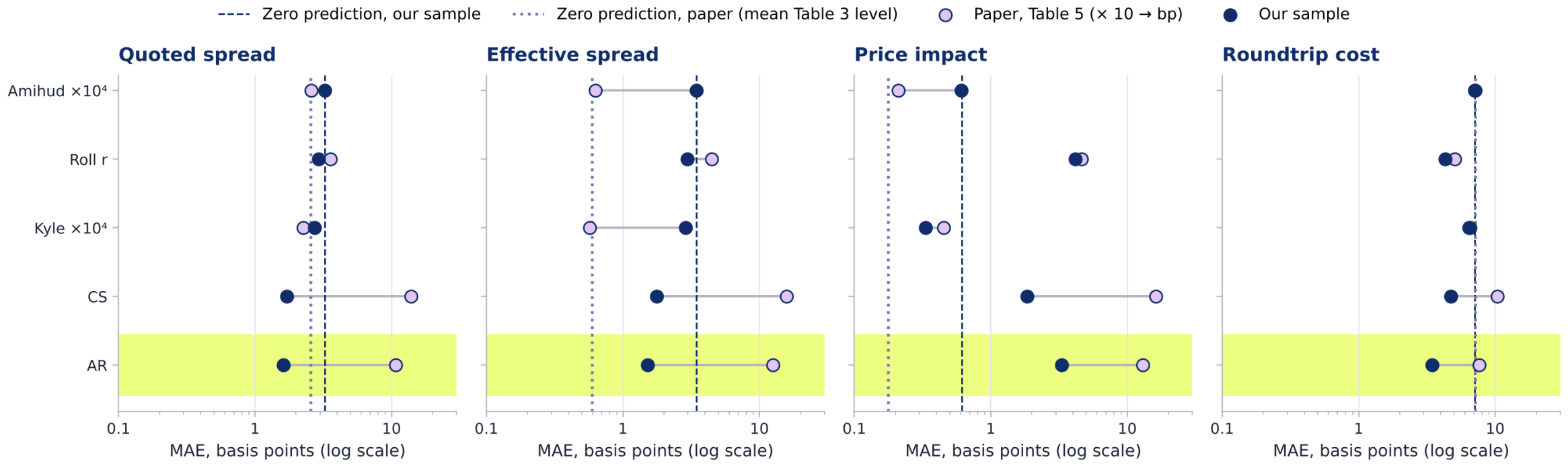
TX with QS: −0.21 calm, +0.29 volatile

Terciles of hourly volatility per market; Pearson per venue, averaged over the three venues. Paper values from its Figure 6. Return and volume splits: notebook Section 7.3.

EXTENSION · PART 4

Level accuracy needs a baseline

Mean absolute error against each benchmark (Bitcoin, bp, log scale), with the error of always predicting zero.



Paper: Kyle and Amihud are the most accurate in levels.

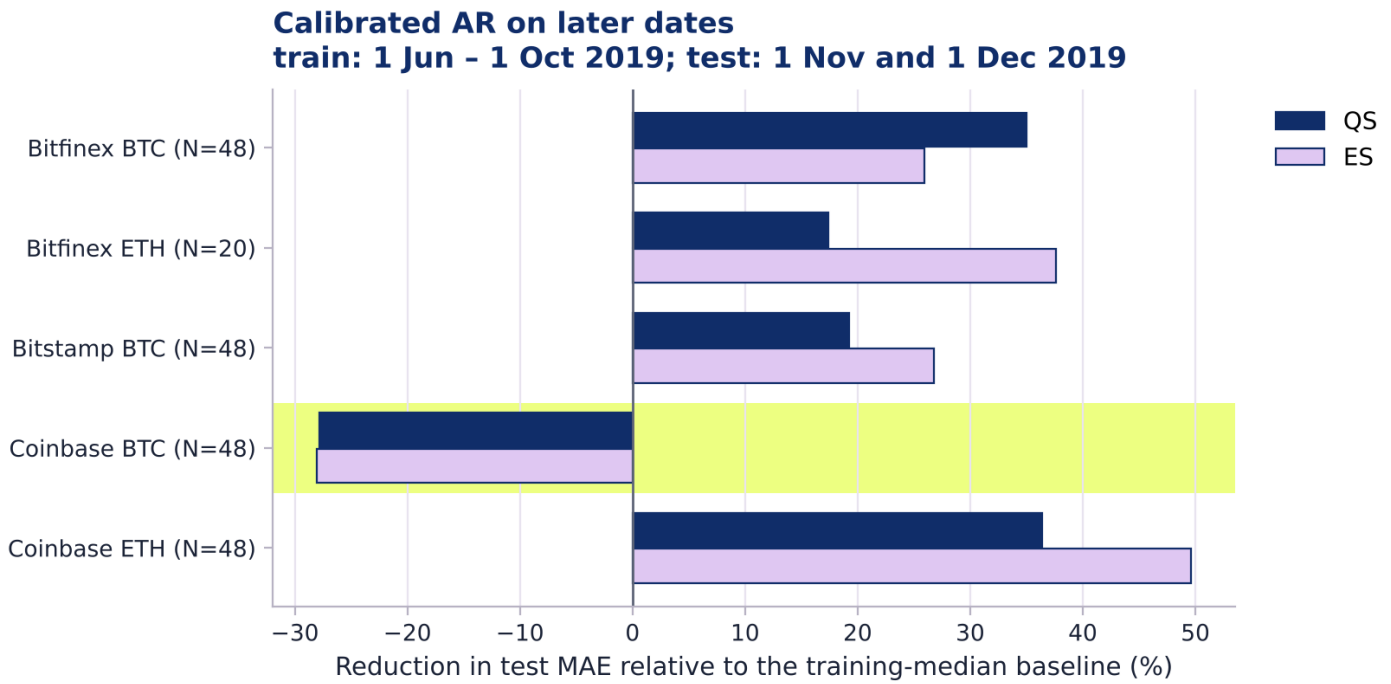
Ours: Amihud's error = predicting zero (QS 3.25 vs 3.25 bp; paper 2.58 vs ≈ 2.54). AR has the smallest QS and ES errors: 1.61 and 1.52 bp.

Mean of three venues, hourly. Paper: Table 5 printed ×1,000, multiplied by 10 (our reading of its units). Dotted line: the paper's mean level = error of predicting zero (QS, CRT; lower bound for ES, PI).

EXTENSION · PART 4

Does a calibrated AR measure costs on later dates?

Train on 1 Jun – 1 Oct 2019, test on 1 Nov and 1 Dec 2019, compare with the training median.



1. Fit benchmark = a + b × proxy on training dates only
2. Apply it to the same hours on the test dates
3. Compare with zero and training-median baselines

QS +17–36% · ES +26–50%
lower test error in 4 of 5 markets

Coinbase BTC −28%
Test-day spreads fell below the training level; predicting zero beats every calibrated proxy.

Calibration fixes AR's level bias only while the cost level is stable. Raw AR: 1.2–3.1× the baseline's QS error.

Affine calibration of the standardized proxy, fitted on training dates only; QS and ES clipped at 0. Measures the same hour, not a forecast. Bitstamp ETH omitted (1 test hour).

SYNTHESIS

Which findings of the paper replicate?

Bitcoin, hourly. Lilac column: the paper's numbers; navy column: our sample.

Finding of the paper	Paper	Our sample	Verdict
AR best for QS, ES, CRT	0.76 / 0.84 / 0.71	0.75 / 0.80 / 0.68	Replicated
AR best for PI	0.59	0.58 ($Vol 0.60)	Partly
CS second	for QS, ES, PI	for QS, ES, CRT25	Largely
Amihud uncorrelated	−0.01 to 0.03	−0.20 to 0.06	Replicated
TX, $Vol positive with costs	0.35 to 0.72	0.41 to 0.60	Replicated*
Kyle tracks the roundtrip cost	0.69	0.51	Weaker
AR best in calm and volatile hours	yes	yes for QS and ES	Largely
Correlations higher in volatile hours	yes	no: lower for AR	Not replicated
Kyle and Amihud best in levels	yes	Amihud = zero baseline	Not replicated
AR ranks venues best (QS)	66.67%	75.9%	Replicated
Benchmark levels (Table 3)	–	QS, CRT close; ES, PI 2.4–8.5×	Partly

* After removing volatility, the activity–cost link mostly disappears (slide 15). Full table: notebook Section 6.7.

NOVELTY · OUR CONTRIBUTION

Our contribution: what we add beyond the paper

All six points are implemented and run in our final notebook (Sections 3–7).

1 Independent data

The six markets rebuilt from raw Tardis.dev trades, quotes and 25-level books, mostly after the paper's period, with unit and integration tests.

2 Volatility control

Partial Spearman correlations: AR's link to spreads survives, its link to price impact does not; the activity–cost sign is largely volatility.

3 Uncertainty for 7 days

Date-block bootstrap with 999 resamples of whole days: 95% intervals for all 192 partial correlations.

4 A baseline for levels

Errors compared with predicting zero: Amihud's "accuracy" equals the zero forecast, in our data and in the paper's.

5 Out-of-sample calibration

Train June–October, test November–December against zero and median baselines: helps in 4 of 5 markets, fails when the level shifts.

6 Transparent definitions

ES absolute and signed, signed PI, Table 5 units documented; checks of quote age (30 vs 60 s), book depth and proxy bias by simulation.

Planned but not done: the Binance 2024 → 2025 extension from our project plan (see Limitations).

CRITIQUE

Limitations: what our design cannot show

Seven selected days: our findings are indicative rather than conclusive.

Sample
7 first-of-month days (807 hours vs ≈ 9,500 in the paper); Bitstamp ETH only 28 hours; recording gaps on 1 Aug 2019.

Different data
WebSocket recordings on a 10-second grid instead of REST snapshots: explains part of the higher ES and PI levels.

Book depth
25 instead of 50 levels: CRT25 approximates CRT, although 25 levels fill the order in 99.85% of snapshots.

Scope
Hourly only: no daily or 15-day frequency, quantile dependence or PCA composite; the planned Binance 2024 → 2025 extension was not done.

Definitions
ES, PI and Table 5 units follow our documented reading of the SSRN text; the authors have not confirmed it.

Reproducibility
FAST_MODE loads ES, PI and CRT25 from the full run; only the full mode (several hours) recomputes them from raw data.

Notebook Section 8.4.

CONCLUSIONS

Conclusions

What we can say after replicating and extending the paper.

01	**The central result travels**	On independent 2019 data, AR again tracks spreads and roundtrip costs best and ranks the venues best.
02	**Volatility is a common driver**	The positive activity–cost link mostly disappears once volatility is removed; AR keeps 0.41–0.61 with QS.
03	**Price impact is the hardest**	No proxy tracks it after the volatility control, and AR ranks the venues by it no better than chance.
04	**Levels need a baseline**	Amihud's small error is the zero forecast; AR is biased upwards at minute frequency.
05	**For practice**	Use AR to compare venues and time liquidity; for cost levels, calibrate on recent data and check against a simple baseline.

REFERENCES

References, data and tools

Main paper

Brauneis, A., Mestel, R., Riordan, R., Theissen, E. (2021). How to measure the liquidity of cryptocurrency markets? Journal of Banking & Finance 124, 106041. We worked with the SSRN version of March 2020.

Proxy methods

Roll (1984); Amihud (2002); Corwin & Schultz (2012); Kyle & Obizhaeva (2016); Abdi & Ranaldo (2017), as cited in the paper.

Data

Tardis.dev historical CSV datasets: trades, quotes and book_snapshot_25 for Bitfinex, Bitstamp and Coinbase, seven days of 2019.

Code

Group17_Cryptocurrency_Liquidity_Measurement_notebook.ipynb (Python 3, pandas, NumPy, matplotlib). FAST_MODE about one minute; full mode several hours.

Use of language models: an LLM (Claude, Anthropic) helped us implement and review code and draft explanations and slides. All numbers come from the notebook's computed outputs.

Thank you

Questions are welcome



...and may your spreads always be tight

Chistiakova Maria · Rudko Sofia · Meyzler Liya · Samoilova Maria

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BACKUP · PART 2

Proxy details for questions

Short answers on the eight proxies (Sofia).

Why average TX over minutes with trades?

That is the paper's definition (Section 2.2.2): a minute without trades is a gap, not a zero.

Does setting negative pairs to 0 bias AR and CS?

Yes, upwards, as in the paper. 29–60% of pairs are negative; “average first, clip after” gives CS = 0 in five of six markets.

Why is Roll so often zero?

Spreads below 1 bp against minute moves of about 7 bp: the covariance is often positive, so Roll = 0. The simulation shows the same.

Why does AR keep the square root?

As in the SSRN formula: AR is then a spread in bp, comparable with QS.

What are Kyle's units?

Kyle divides by total hourly USD volume, so its level depends on units; we scale it $\times 10^4$ only for level comparisons.

What about odd Bitfinex prices?

33.4% of BTC and 21.5% of ETH minutes have sub-cent prices; kept as recorded and flagged.

Notebook Sections 4.6–4.9.